

# Thermo Scientific™ DynaDrive™ Single-use Bioreactor 5000 L - 250-5000 L working volume

thermo-dynadrive-5000

## Field-level manufacturer evidence; remaining values unverified

Each field is classified independently. Direct manufacturer values take precedence over estimates only for the matching model and configuration. Approximate values printed by the manufacturer remain approximate. Unverified fields are editable simulation assumptions, not equipment operating limits.

| Parameter                                   | Value in simulator    | Evidence status     |
|---------------------------------------------|-----------------------|---------------------|
| Minimum working volume                      | 250 L                 | Source-supported    |
| Maximum working volume                      | 5000 L                | Source-supported    |
| Nominal vessel / bag size                   | 5000 L                | Source-supported    |
| Total vessel volume loaded into the form    | 5585 L                | Source-supported    |
| Initial working volume loaded into the form | 250 L                 | Derived / estimated |
| Product-contact material                    | Plastic / single-use  | App assumption      |
| Height : diameter                           | 2.5                   | Derived / estimated |
| Internal / chamber diameter                 | 1.37 m                | Source-supported    |
| Liquid height at maximum working volume     | 3.42 m                | Source-supported    |
| Impeller / mixing type                      | Pitched-blade turbine | App assumption      |
| Impeller count                              | 5                     | Source-supported    |
| Impeller diameter                           | 0.49 m                | Source-supported    |
| Power number                                | 0.4                   | App assumption      |
| Initial agitation                           | 55 rpm                | App assumption      |
| Maximum agitation                           | 85 rpm                | App assumption      |
| Initial aeration                            | 0.05 vvm              | App assumption      |
| Maximum aeration                            | 0.3 vvm               | App assumption      |
| Baffled vessel                              | No                    | App assumption      |
| Maximum pressure                            | 0.034 bar(g)          | Source-supported    |
| Default operating pressure                  | 0 bar(g)              | App assumption      |
| Maximum inlet oxygen fraction               | 100%                  | App assumption      |

## Parameter evidence and derivations

### Minimum working volume - 250 L

Source-supported | Thermo DynaDrive 3000 and 5000 L unpacking and setup guide (Spanish) - PDF page 48; Table 4.1, 5000 L column; printed p. 46

Rule: Direct manufacturer specification; no reverse calculation.

Selection / calculation: Selected value: 250 L.

Limitations: DOC0176ES Rev. A, February 2022. Spanish original. Direct metric dimensions; 137 cm chamber and 49 cm impeller span both columns. Hardware combines modified two-blade impellers with one two-blade sweep impeller; count totals the assemblies, not blades.

Local original: [Thermo DynaDrive 3000 and 5000 L unpacking and setup guide \(Spanish\) - PDF page 48](#)

[Manufacturer publication](#)

### Maximum working volume - 5000 L

Source-supported | Thermo DynaDrive 3000 and 5000 L unpacking and setup guide (Spanish) - PDF page 48; Table 4.1, 5000 L column; printed p. 46

Rule: Direct manufacturer specification; no reverse calculation.

Selection / calculation: Selected value: 5000 L.

Limitations: DOC0176ES Rev. A, February 2022. Spanish original. Direct metric dimensions; 137 cm chamber and 49 cm impeller span both columns. Hardware combines modified two-blade impellers with one two-blade sweep impeller; count totals the assemblies, not blades.

Local original: [Thermo DynaDrive 3000 and 5000 L unpacking and setup guide \(Spanish\) - PDF page 48](#)

[Manufacturer publication](#)

### Nominal vessel / bag size - 5000 L

Source-supported | Thermo DynaDrive 3000 and 5000 L unpacking and setup guide (Spanish) - PDF page 48; Table 4.1, 5000 L column; printed p. 46

Rule: Direct manufacturer specification; no reverse calculation.

Selection / calculation: Selected value: 5000 L.

Limitations: DOC0176ES Rev. A, February 2022. Spanish original. Direct metric dimensions; 137 cm chamber and 49 cm impeller span both columns. Hardware combines modified two-blade impellers with one two-blade sweep impeller; count totals the assemblies, not blades.

Local original: [Thermo DynaDrive 3000 and 5000 L unpacking and setup guide \(Spanish\) - PDF page 48](#)

[Manufacturer publication](#)

### Total vessel volume loaded into the form - 5585 L

Source-supported | Thermo DynaDrive 3000 and 5000 L unpacking and setup guide (Spanish) - PDF page 48; Table 4.1, 5000 L column; printed p. 46

Rule: Direct manufacturer specification; no reverse calculation.

Selection / calculation: Selected value: 5585 L.

Limitations: DOC0176ES Rev. A, February 2022. Spanish original. Direct metric dimensions; 137 cm chamber and 49 cm impeller span both columns. Hardware combines modified two-blade impellers with one two-blade sweep impeller; count totals the assemblies, not blades.

Local original: [Thermo DynaDrive 3000 and 5000 L unpacking and setup guide \(Spanish\) - PDF page 48](#)

[Manufacturer publication](#)

### Initial working volume loaded into the form - 250 L

Derived / estimated | Form-population rule

Rule: initial working volume = min(minimum working volume, maximum working volume)

Selection / calculation: min(250 L, 5000 L) = 250 L

Limitations: The preset starts the process at the lower end of its configured working range.

## Product-contact material - Plastic / single-use

App assumption | Unverified legacy preset

Rule: Retained or selected modelling value; not a verified hardware limit.

Selection / calculation: Selected value: Plastic / single-use.

Limitations: No matching model/configuration-specific primary specification was verified for this field. Legacy values remain editable modelling assumptions, not equipment ratings. Family marketing ranges, external dimensions, sensor ranges, jacket ratings and experimental setpoints are not substituted for vessel specifications.

## Height : diameter - 2.5

Derived / estimated | Thermo DynaDrive 3000 and 5000 L unpacking and setup guide (Spanish) - PDF page 48; Table 4.1 liquid height and chamber diameter

Rule: liquid H/D = published liquid height / published chamber diameter

Selection / calculation:  $3.42 / 1.37 = 2.4963503649635035$

Limitations: Published fluid ratios 1.7/2.9 do not match the absolute dimensions. Keep direct heights/diameter, calculate their quotient, and preserve the inconsistency.

Local original: [Thermo DynaDrive 3000 and 5000 L unpacking and setup guide \(Spanish\) - PDF page 48](#)

Manufacturer publication

## Internal / chamber diameter - 1.37 m

Source-supported | Thermo DynaDrive 3000 and 5000 L unpacking and setup guide (Spanish) - PDF page 48; Table 4.1, 5000 L column; printed p. 46

Rule: Direct manufacturer specification; no reverse calculation.

Selection / calculation: Selected value: 1.37 m.

Limitations: DOC0176ES Rev. A, February 2022. Spanish original. Direct metric dimensions; 137 cm chamber and 49 cm impeller span both columns. Hardware combines modified two-blade impellers with one two-blade sweep impeller; count totals the assemblies, not blades.

Local original: [Thermo DynaDrive 3000 and 5000 L unpacking and setup guide \(Spanish\) - PDF page 48](#)

Manufacturer publication

## Liquid height at maximum working volume - 3.42 m

Source-supported | Thermo DynaDrive 3000 and 5000 L unpacking and setup guide (Spanish) - PDF page 48; Table 4.1, 5000 L column; printed p. 46

Rule: Direct manufacturer specification; no reverse calculation.

Selection / calculation: Selected value: 3.42 m.

Limitations: DOC0176ES Rev. A, February 2022. Spanish original. Direct metric dimensions; 137 cm chamber and 49 cm impeller span both columns. Hardware combines modified two-blade impellers with one two-blade sweep impeller; count totals the assemblies, not blades.

Local original: [Thermo DynaDrive 3000 and 5000 L unpacking and setup guide \(Spanish\) - PDF page 48](#)

Manufacturer publication

## Impeller / mixing type - Pitched-blade turbine

App assumption | Thermo DynaDrive 3000 and 5000 L unpacking and setup guide (Spanish) - PDF page 48; Table 4.1 Propulsor

Rule: Retained or selected modelling value; not a verified hardware limit.

Selection / calculation: Selected value: Pitched-blade turbine.

Limitations: Mixed modified pitched-blade plus sweep arrangement cannot be represented exactly by the generic single-type simulator. Generic Np remains unverified.

Local original: [Thermo DynaDrive 3000 and 5000 L unpacking and setup guide \(Spanish\) - PDF page 48](#)

Manufacturer publication

## Impeller count - 5

Source-supported | Thermo DynaDrive 3000 and 5000 L unpacking and setup guide (Spanish) - PDF page 48; Table 4.1, 5000 L column; printed p. 46

Rule: Direct manufacturer specification; no reverse calculation.

Selection / calculation: Selected value: 5.

Limitations: DOC0176ES Rev. A, February 2022. Spanish original. Direct metric dimensions; 137 cm chamber and 49 cm impeller span both columns. Hardware combines modified two-blade impellers with one two-blade sweep impeller; count totals the assemblies, not blades.

Local original: [Thermo DynaDrive 3000 and 5000 L unpacking and setup guide \(Spanish\) - PDF page 48](#)

[Manufacturer publication](#)

## Impeller diameter - 0.49 m

Source-supported | Thermo DynaDrive 3000 and 5000 L unpacking and setup guide (Spanish) - PDF page 48; Table 4.1, 5000 L column; printed p. 46

Rule: Direct manufacturer specification; no reverse calculation.

Selection / calculation: Selected value: 0.49 m.

Limitations: DOC0176ES Rev. A, February 2022. Spanish original. Direct metric dimensions; 137 cm chamber and 49 cm impeller span both columns. Hardware combines modified two-blade impellers with one two-blade sweep impeller; count totals the assemblies, not blades.

Local original: [Thermo DynaDrive 3000 and 5000 L unpacking and setup guide \(Spanish\) - PDF page 48](#)

[Manufacturer publication](#)

## Power number - 0.4

App assumption | Generic mammalian stirred-tank profile

Rule: `PROFILE_DEFAULTS["mammalian"].powerNumber = 0.4`

Selection / calculation: No calculation. The vessel inherits this literal value from its generic profile.

Limitations: No model-specific value is stored for this parameter, so the simulator uses its editable mammalian stirred-tank starting profile. The power number is an empirical generic value used in  $P = N_p N^3 D_i^5$ . It is not recalculated from blade geometry.

## Initial agitation - 55 rpm

App assumption | Large mammalian-vessel scaling rule

Rule:  $\text{baseRpm} = \text{maxVolume} \geq 1000 \text{ L} ? 55 : 80$

Selection / calculation: 5000 L is at least 1000 L -> 55 rpm

Limitations: The simulator lowers its generic starting agitation at larger scale. This is an internal scale heuristic, not a manufacturer limit. This is the initial agitation setting loaded into the form.

## Maximum agitation - 85 rpm

App assumption | Thermo DynaDrive 3000 and 5000 L unpacking and setup guide (Spanish) - PDF page 48; Table 4.1 agitation; compare Table 4.2 p. 49

Rule: Retained or selected modelling value; not a verified hardware limit.

Selection / calculation: Selected value: 85 rpm.

Limitations: Table 4.1 gives VFD range 0-85 rpm; Table 4.2 on p. 49 gives motor speed 0-90 rpm. Lower ceiling chosen pending clarification. Recommended culture minimum is >30 rpm, not a guaranteed safe range for all conditions.

Local original: [Thermo DynaDrive 3000 and 5000 L unpacking and setup guide \(Spanish\) - PDF page 48](#)

[Manufacturer publication](#)

## Initial aeration - 0.05 vvm

App assumption | Generic mammalian stirred-tank profile

Rule: `PROFILE_DEFAULTS["mammalian"].baseVvm = 0.05`

Selection / calculation: No calculation. The vessel inherits this literal value from its generic profile.

Limitations: No model-specific value is stored for this parameter, so the simulator uses its editable mammalian stirred-tank starting profile. This is the initial gas rate loaded into the form.

## Maximum aeration - 0.3 vvm

App assumption | Generic mammalian stirred-tank profile

Rule: PROFILE\_DEFAULTS["mammalian"].maxVvm = 0.3

Selection / calculation: No calculation. The vessel inherits this literal value from its generic profile.

Limitations: No model-specific value is stored for this parameter, so the simulator uses its editable mammalian stirred-tank starting profile. This is the gas-rate ceiling available to the generic DO-control cascade.

## Baffled vessel - No

App assumption | Generic mammalian stirred-tank profile

Rule: PROFILE\_DEFAULTS["mammalian"].baffled = false

Selection / calculation: No calculation. The vessel inherits this literal value from its generic profile.

Limitations: No model-specific value is stored for this parameter, so the simulator uses its editable mammalian stirred-tank starting profile. This Boolean changes the generic vessel configuration and mixing interpretation.

## Maximum pressure - 0.034 bar(g)

Source-supported | Thermo DynaDrive 3000 and 5000 L unpacking and setup guide (Spanish) - PDF page 49; Table 4.2 Presion maxima de la bolsa

Rule: Direct manufacturer specification; no reverse calculation.

Selection / calculation: Selected value: 0.034 bar(g).

Limitations: Direct maximum bag pressure 34 mbar (0.5 psi), converted to 0.034 bar; not condenser TCU or jacket pressure.

Local original: [Thermo DynaDrive 3000 and 5000 L unpacking and setup guide \(Spanish\) - PDF page 49](#)

Manufacturer publication

## Default operating pressure - 0 bar(g)

App assumption | Thermo DynaDrive 3000 and 5000 L unpacking and setup guide (Spanish) - PDF page 49; Table 4.2 bag pressure

Rule: Retained or selected modelling value; not a verified hardware limit.

Selection / calculation: Selected value: 0 bar(g).

Limitations: Atmospheric app starting value replaces 0.05 bar, above the published bag limit; not a recommended culture setpoint.

Local original: [Thermo DynaDrive 3000 and 5000 L unpacking and setup guide \(Spanish\) - PDF page 49](#)

Manufacturer publication

## Maximum inlet oxygen fraction - 100%

App assumption | Fixed simulator capability assumption

Rule: maxOxygenFraction = 100

Selection / calculation: No calculation. Every vessel preset currently permits a 100% oxygen inlet ceiling in the generic gas cascade.

Limitations: This describes the simulator's permitted gas-mixture range, not proof that the selected hardware is certified for pure oxygen service.

## Configuration notes and unresolved evidence

DynaDrive is represented with a generic mammalian stirred-tank profile unless model-specific values are explicitly stored.

2026-09-11 field-level review: use the evidence beside each parameter; an archived document does not verify the whole preset.

No matching model/configuration-specific primary specification was verified for this field. Legacy values remain editable modelling assumptions, not equipment ratings. Family marketing ranges, external dimensions, sensor ranges, jacket ratings and experimental setpoints are not substituted for vessel specifications.

## Local original document identities

Thermo DynaDrive 3000 and 5000 L unpacking and setup guide (Spanish)

thermo-dynadrive-3000-5000-setup-es-bed67ef725fc.pdf

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